



Molecular Data for Phylogenetics and Macroevolutionary Study

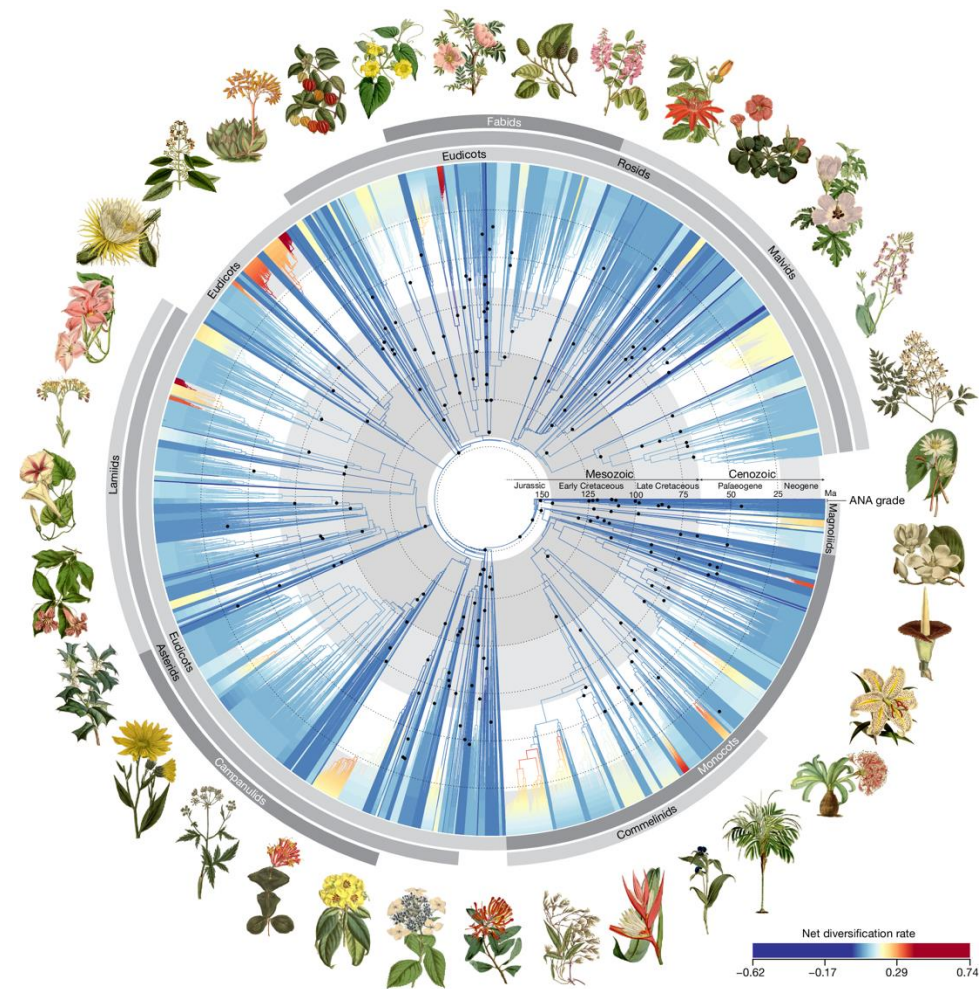
Arong Luo

July 18th, 2025

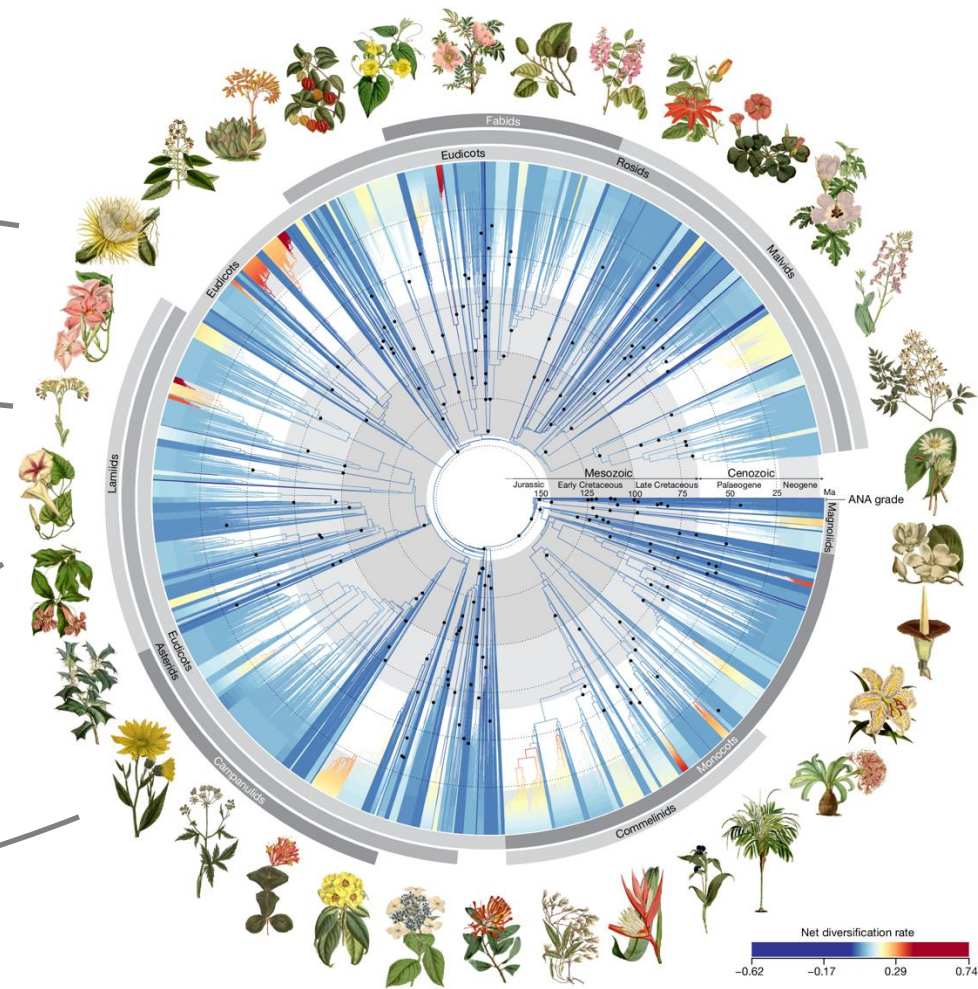
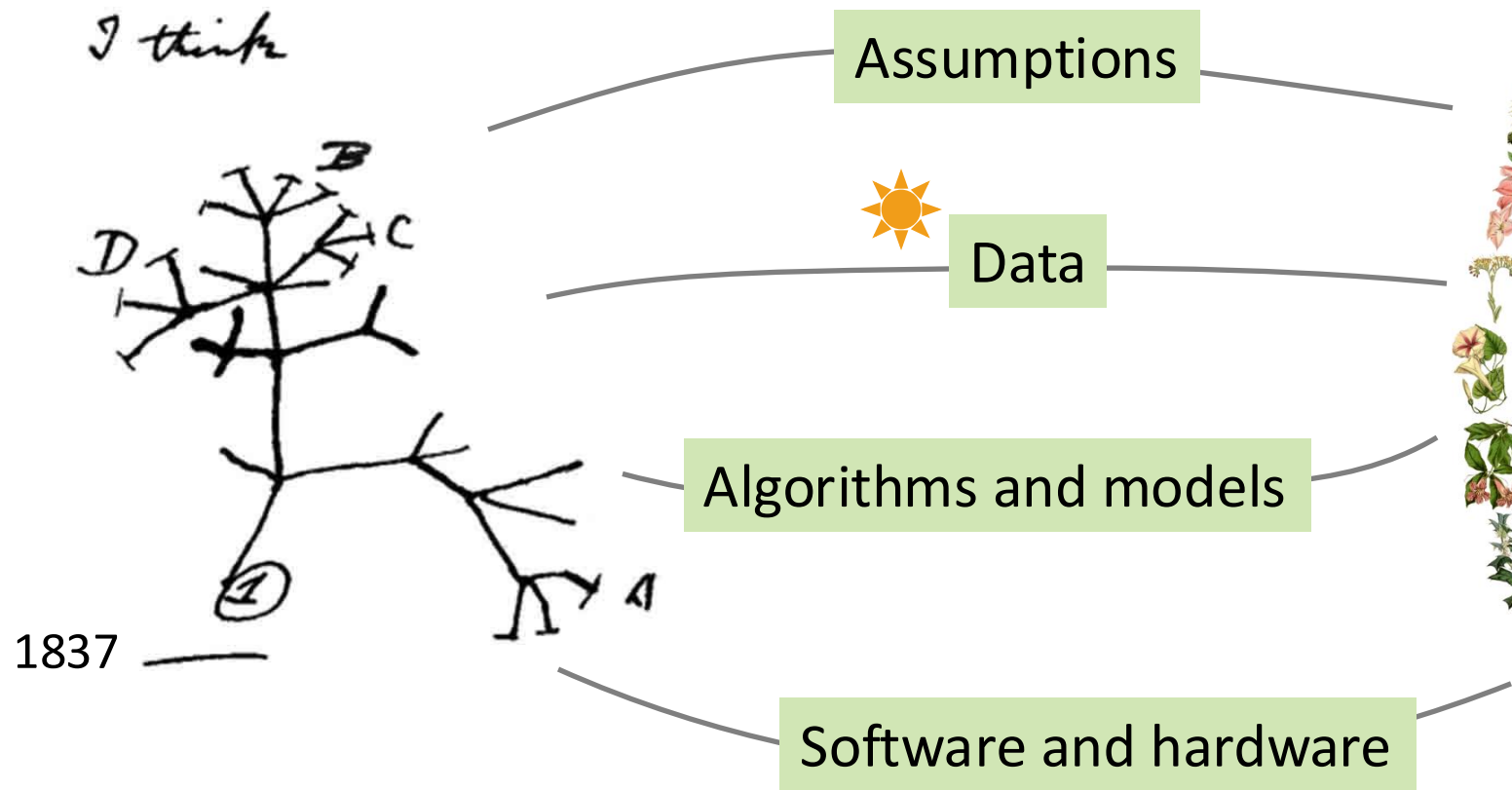
I think



Darwin 1837



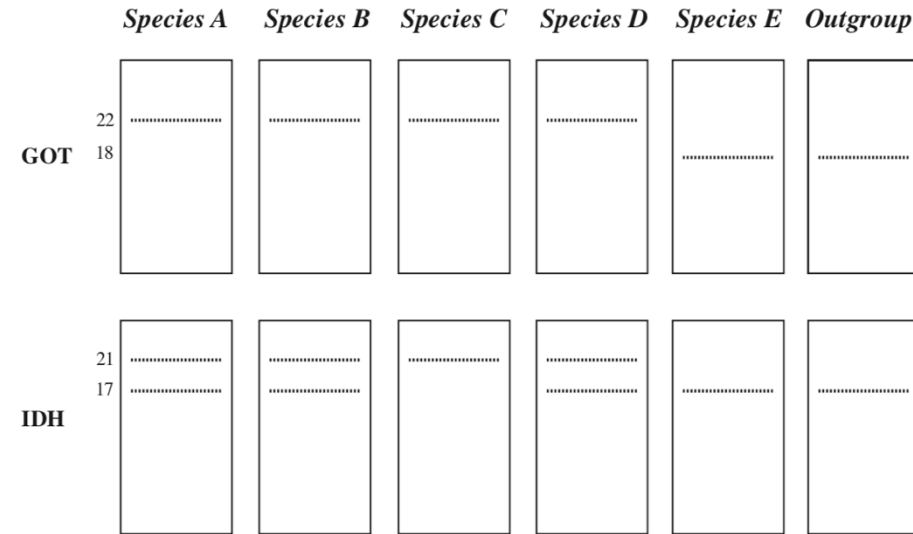
Zuntini *et al.* 2024



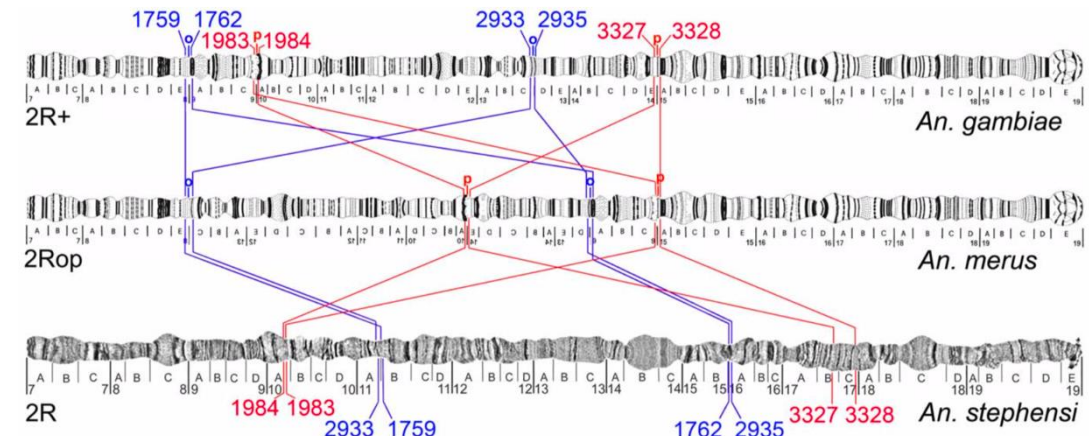
Molecular Data

Molecular data types

- **Allozymes:** different molecular forms of an enzyme; corresponding to different alleles of a common gene (*locus*)



- **Chromosomal inversions:** a chromosome rearrangement; rearrangement or breakpoint distance for analyses



Molecular data types

- **Sequence data:** widely used
 - Nucleotides or amino acids
 - Nuclear, mitochondrial, chloroplast
- **Microsatellites**
- **Single-nucleotide polymorphisms (SNPs)**
- **Restriction fragment length polymorphisms (RFLPs)**
- **Random amplified polymorphic DNA (RAPDs)**
- ...

Taxon 1	G CCTAGCC A AAGCTC T TC C CAAG G T G ACTCTCA G TT C AAGCT
Taxon 2	G CCTAGCC A AAGCTC T TC C CAAG C T G ACTCTCA-----GCT
Taxon 3	G CCTAGCC T AAGCTC A AC C CAAG G T G TCTCTCA G TT C AAGCT
Taxon 4	G CCTAGCC T AAGCTC T TC C CAAG G T G TCTCTCA G TT C AAGCT
Taxon 5	G CCTAGCC A AAGCTC T TC C CAAG C T G ACTCTCA-----GCT
Taxon 6	C CCTAGCC A AAGCTC T TC C CAAG C T G ACTCTCA G TT C AAGCT
Taxon 7	C CCTAGCC A AAGCTC T TC C CAAG C T G ACTCTCA G TT C AAGCT
Taxon 8	G CCTAGCC T AAGCTC T TC C CAAG C T G ACTCTCA G TT C AAGCT

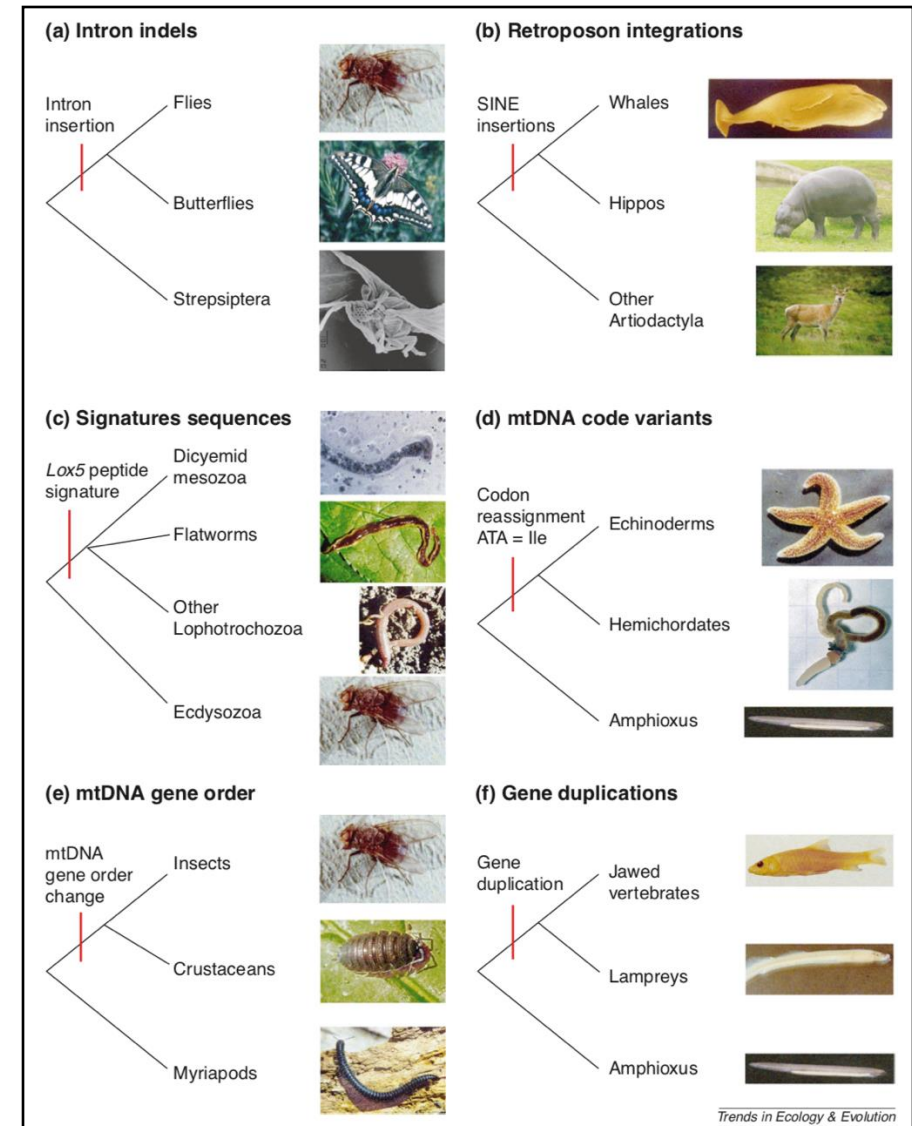
Nucleotide sequences

TAXA	CHARACTERS						
	<i>Eco</i> RI	BAM	BAM	<i>Eco</i> RI	BAM	<i>Eco</i> RI	BAM
	3000	3400	3500	5000	5300	6000	7800
Species A	–	–	+	+	+	+	–
Species B	+	+	–	+	+	+	+

RFLPs

Molecular data types

- **Rare Genomic Changes (RGCs):** large-scale mutational changes that have occurred in the genomes of particular clades
 - Intron indels
 - Retroposon integrations
 - Signatures sequences
 - mtDNA code variants
 - mtDNA gene order
 - Gene duplications
 - microRNA profile
 - ...

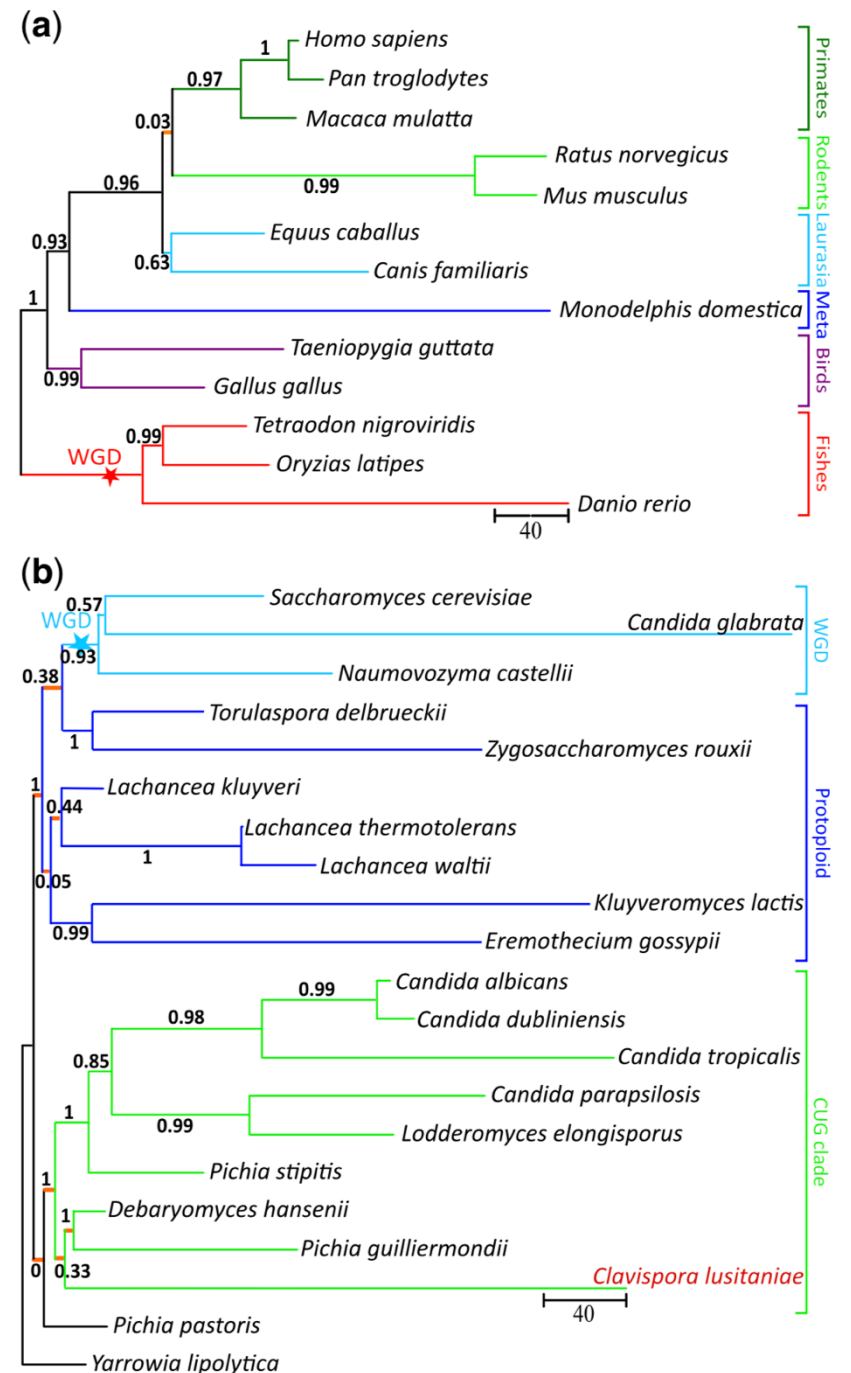


The use of gene order for phylogeny

Table 1. PhyChro Algorithm.

Parts I and II	
for <i>each synteny block</i> do identify the associated pair of breakpoints end for <i>each pair of breakpoints</i> do construct the associated partial splits end while <i># genomes > 3</i> do • construct the Partial Split Distance matrix based on all partial splits; • chose the pair G,H based on minimal distance; • update the partial split list, where G,H are excluded; end for <i>each branch</i> do if <i>the branch is external</i> then compute branch length from trivial partial splits; else compute branch length and confidence score from non trivial partial splits; end end	• Preparation of the input for the iterative step <

Drillon *et al.* 2020

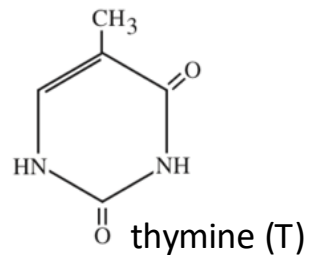
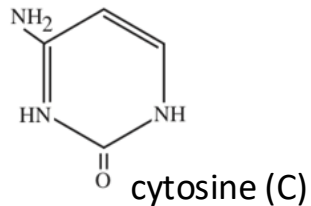
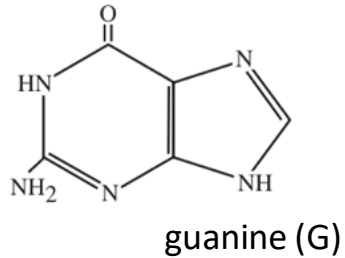
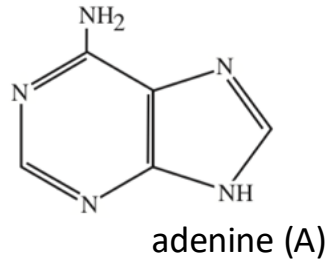


Selecting data

- **Select data to optimize signal: noise**
 - Slowly evolving markers for deep evolutionary events
 - Rapidly evolving markers for recent evolutionary events
- **Minimise violations of phylogenetic assumptions**
 - Some data sets can yield biased phylogenetic estimates
- **Take advantage of existing resources**



Character homology

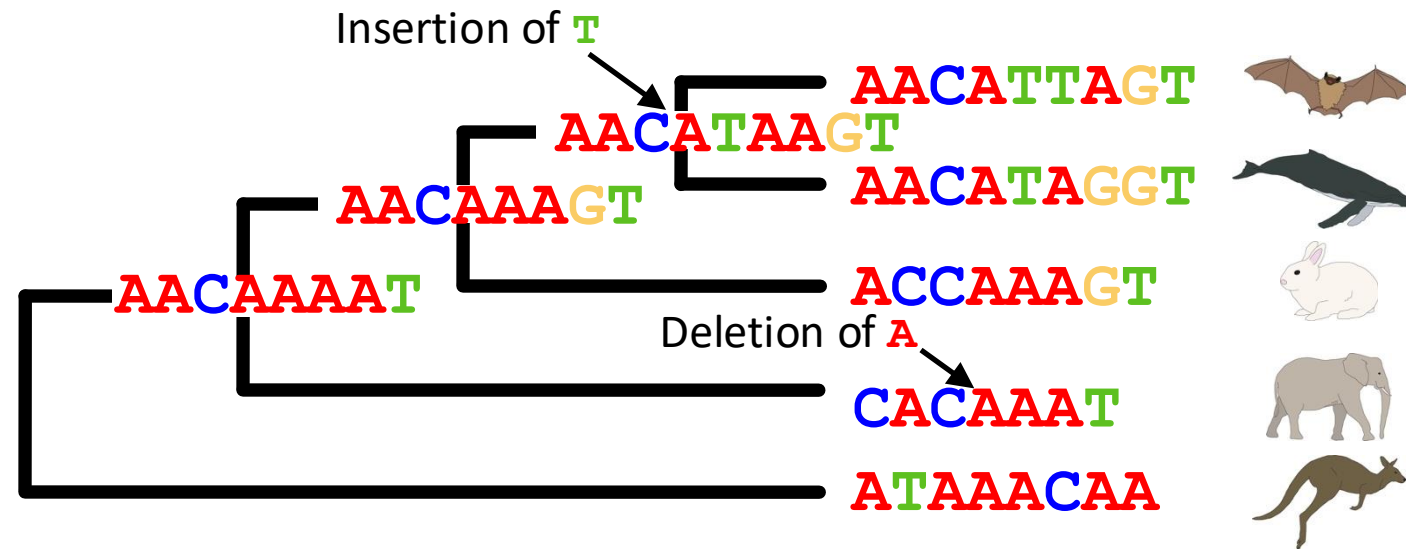


Suppose a **random** sequence from a genome of some taxon.

Given the 4 nucleotides and the length of the sequence n :
there are 4^n possibilities.

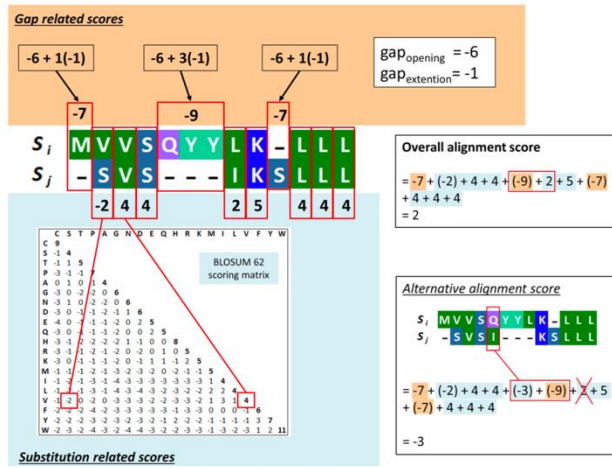
Character homology is one fundamental assumption in the use of sequence data.

Homology: similarity due to shared ancestry, regardless of current differences.

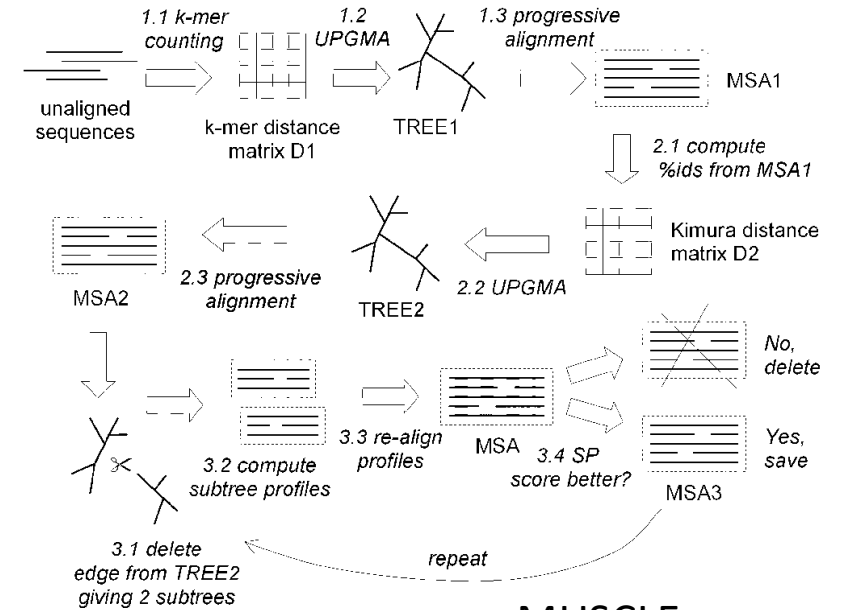
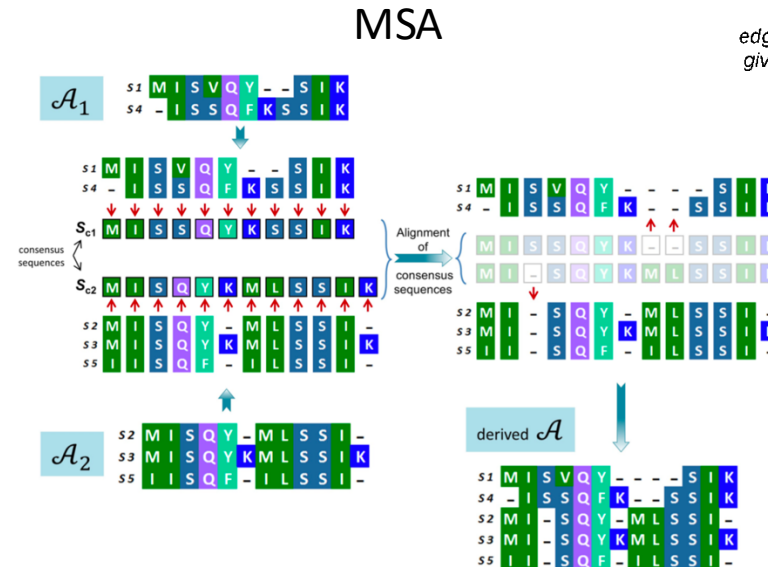


Sequence alignment

The aim of sequence alignment is to maximize the number of sites for which we can infer **homology**.



Pairwise alignment



Sequence alignment

A practical approach: automated methods --- adjusted by eye --- delete uncertain homology

MSA filtering rather than manual curation is applied in genomic data: Gblock, TrimAl, HmmerCleaner, ...

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Nucleic Acids Research, 1994, Vol. 22, No. 22 4673–4680

CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice

Julie D.Thompson, Desmond G.Higgins* and Toby J.Gibson*

MUSCLE: a multiple sequence alignment method with reduced time and space complexity

[Robert C Edgar](#) ✉

MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform 🐼

[Kazutaka Katoh](#) ✉, [Kazuharu Misawa](#), [Kei-ichi Kuma](#), [Takashi Miyata](#)

Data partitioning

Partitioning attempts to achieve variation in rates and patterns of evolution among sites by estimating independent models of evolution for subsites of data.

Biological grounds: genes, codon positions, RNA stems vs loops, hydrophobic vs hydrophilic, ...

Gene A Gene B Gene C

Partitions Tip Dates Site Model Clock Model Priors MCMC

Link Site Models Unlink Site Models Link Clock Models Unlink Clock Models Link Trees Unlink Trees

Name	File	Taxa	Sites	Data Type	Site Model	Clock Model	Tree	...
introns	seq_family	181	1844	nucleotide	introns	clock	tree	☐
rag12	seq_family	181	2702	nucleotide	rag12	clock	tree	☐
rag3	seq_family	181	1351	nucleotide	rag3	clock	tree	☐
rrna	seq_family	181	1591	nucleotide	rrna	clock	tree	☐
mt12	seq_family	181	2216	nucleotide	mt12	clock	tree	☐

Gaps and missing data

- **Delete sites with any missing data**

- Potential loss of information
- Problematic for data supermatrices

- **Treat gaps as unresolved data**

- Gap is simultaneously A, C, G, and T
- Most common approach

- ...

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5
Taxon 1					
Taxon 2					
Taxon 3					
Taxon 4					
Taxon 5					
Taxon 6					

Maximise
gene
sampling

Maximise taxon sampling

Phylogenomic Subsampling and the Search for Phylogenetically Reliable Loci

Nicolás Mongiardino Koch ✉

Molecular Biology and Evolution, Volume 38, Issue 9, September 2021, Pages 4025–4038, <https://doi.org/10.1093/molbev/msab151>

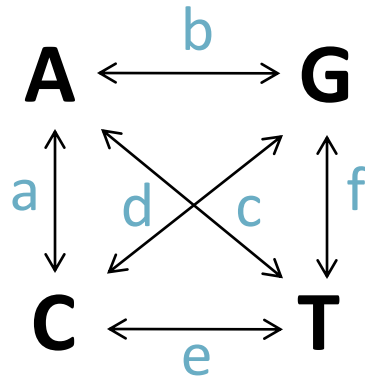
- Impacts of missing data remains **poorly understood**, but can be **important**.

Substitution models

Substitution models: models accounting for the process of sequence evolution, used for inferring genetic distances of taxa.

Nucleotides

Rate Matrix



Base Frequencies

$$\pi_A + \pi_C + \pi_G + \pi_T = 1$$

Site Rates

$$+ I + G$$

JC

$$a=b=c=d=e=f$$

$$\pi_A = \pi_C = \pi_G = \pi_T$$

HKY

$$a=c=d=f, b=e$$

$$\pi_A, \pi_C, \pi_G, \pi_T$$

GTR

$$a, b, c, d, e, f$$

$$\pi_A, \pi_C, \pi_G, \pi_T$$

GTR+I+G

$$a, b, c, d, e, f$$

$$\pi_A, \pi_C, \pi_G, \pi_T$$

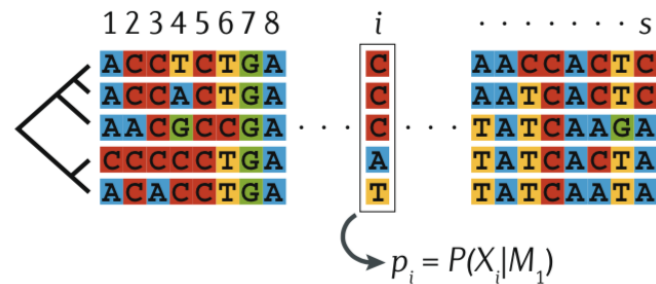
I, G

Substitution models

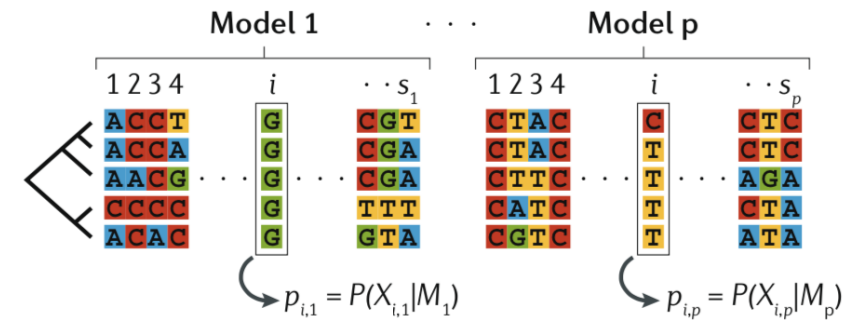
Substitution models: models accounting for the process of sequence evolution, used for inferring genetic distances of taxa.

Nucleotides

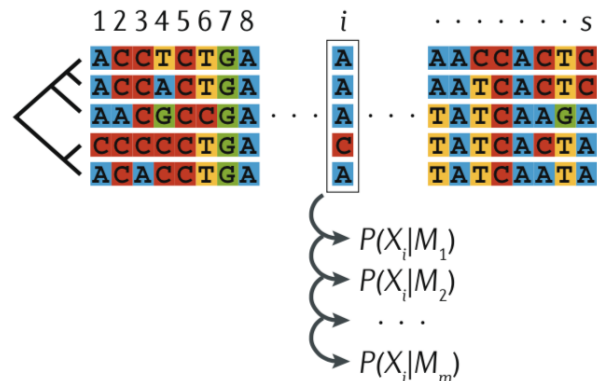
a Homogeneous model



b Partitioned model



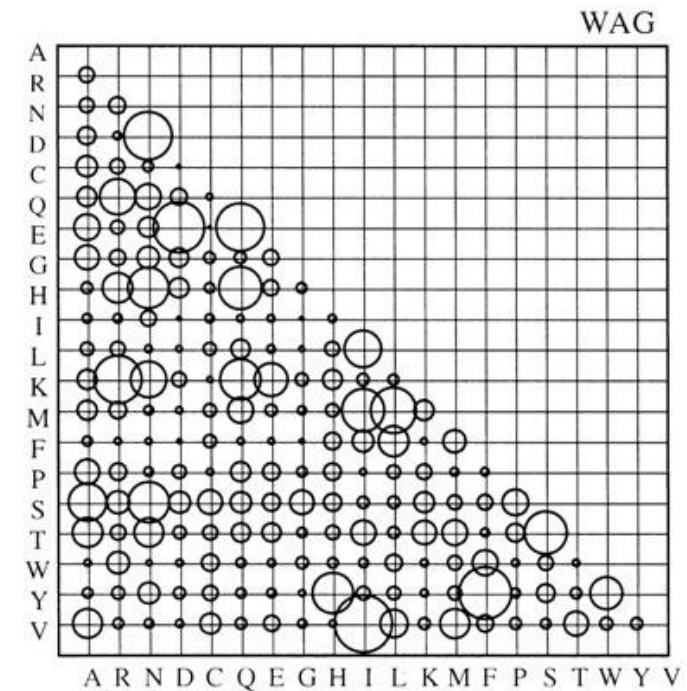
c Mixture model



Substitution models

Amino acids

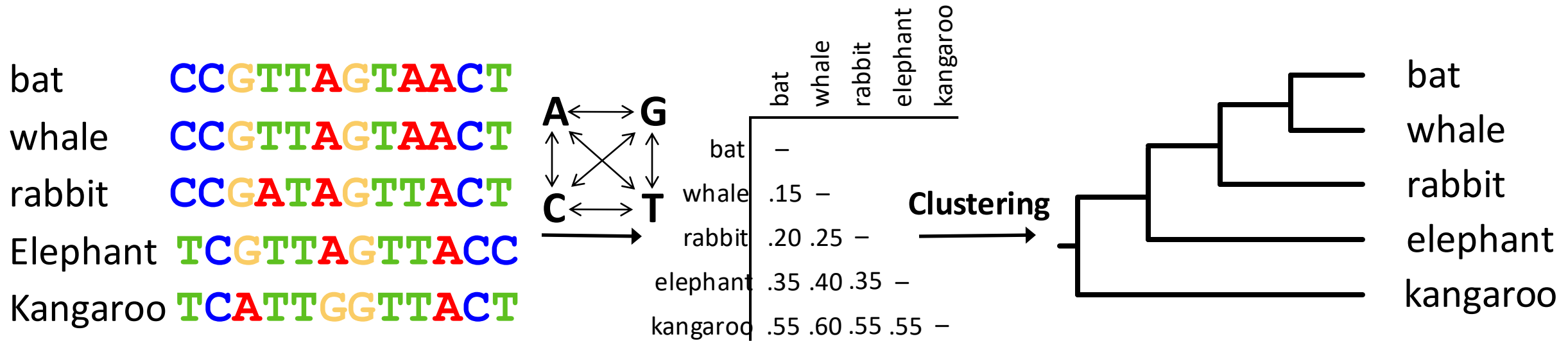
- 20x20 matrix of substitution probabilities
- Too many parameters to estimate
 - GTR model for DNA: 6 parameters
 - GTR model for proteins: 190 parameters
- Estimate substitution probabilities using large data set
 - PAM
 - BLOSUM
 - JTT
 - WAG



Phylogenetic inference

Popular phylogenetic inference methods: maximum parsimony, distance-based methods, maximum likelihood, Bayesian inference

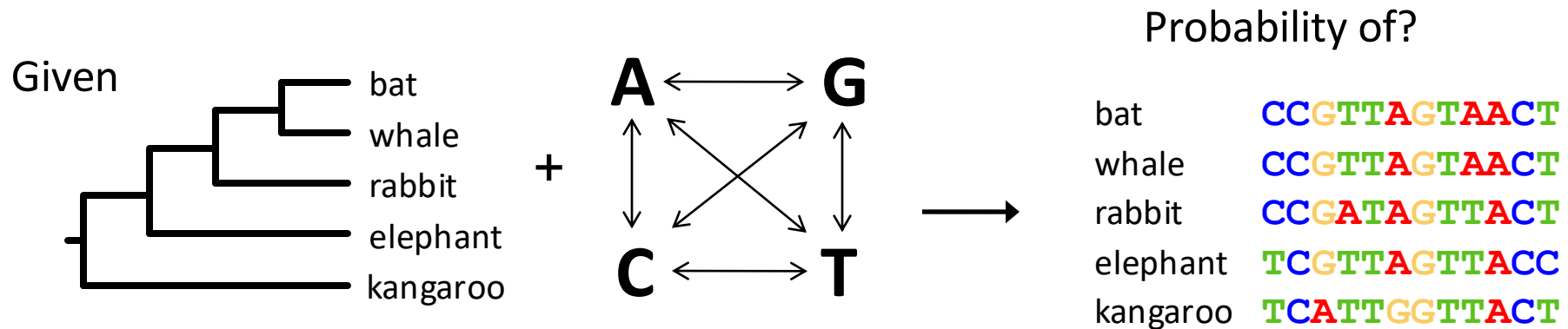
Neighbor joining



Phylogenetic inference

Popular phylogenetic inference methods: maximum parsimony, distance-based methods, maximum likelihood, Bayesian inference

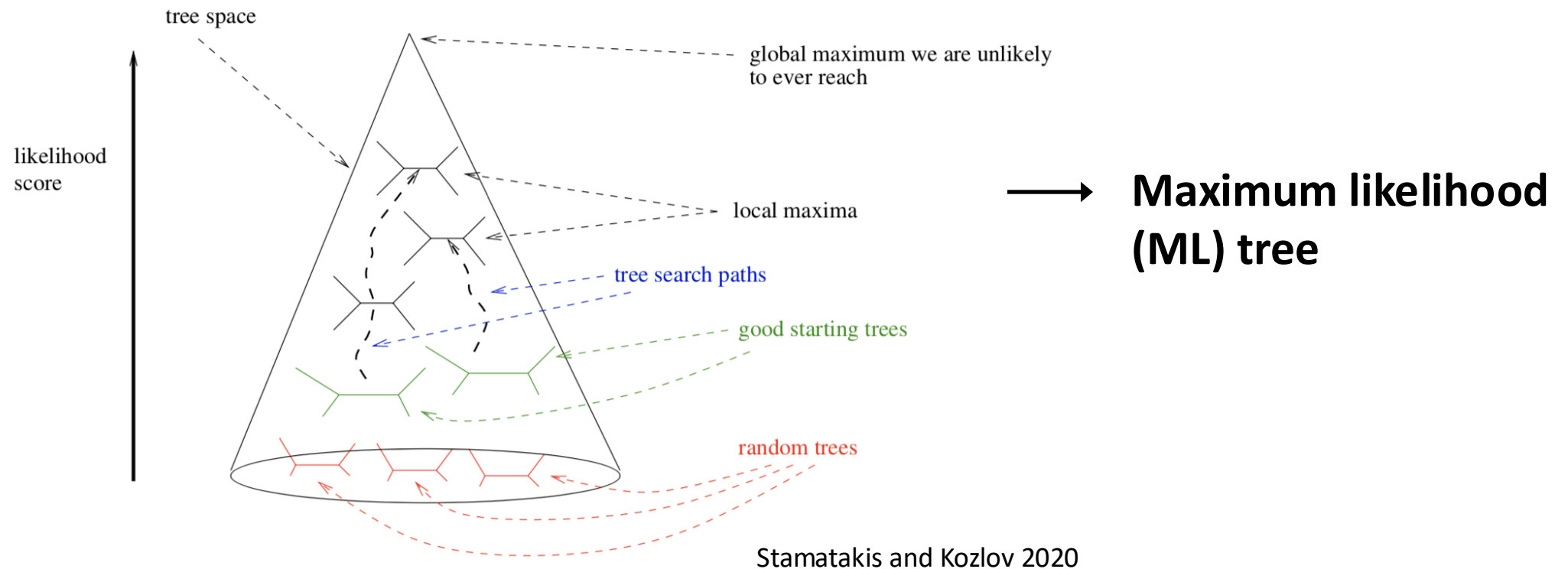
Maximum likelihood



Phylogenetic inference

Popular phylogenetic inference methods: maximum parsimony, distance-based methods, maximum likelihood, Bayesian inference

Maximum likelihood



High-Throughput Data

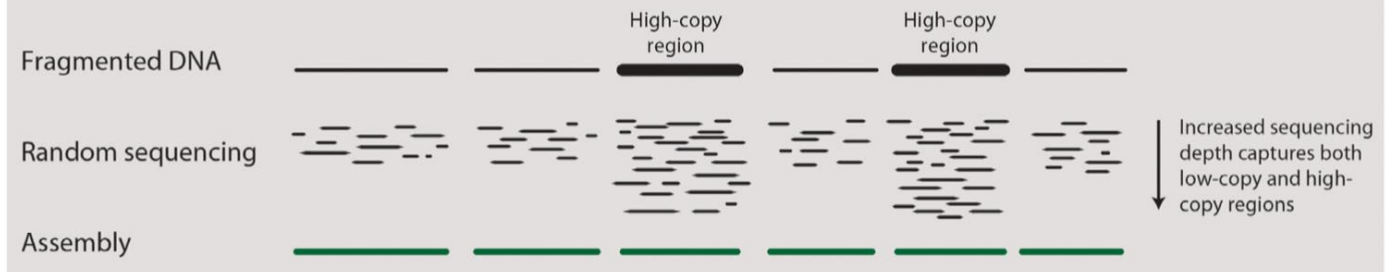
To generate data

- **Shotgun sequencing** involves fragmenting template DNA into short pieces, which are then randomly sequenced to get reads.
 - **Genome skimming**
 - **Whole-genome sequencing**
 - **RNA sequencing**

(A) Genome skimming



(B) Whole-genome sequencing

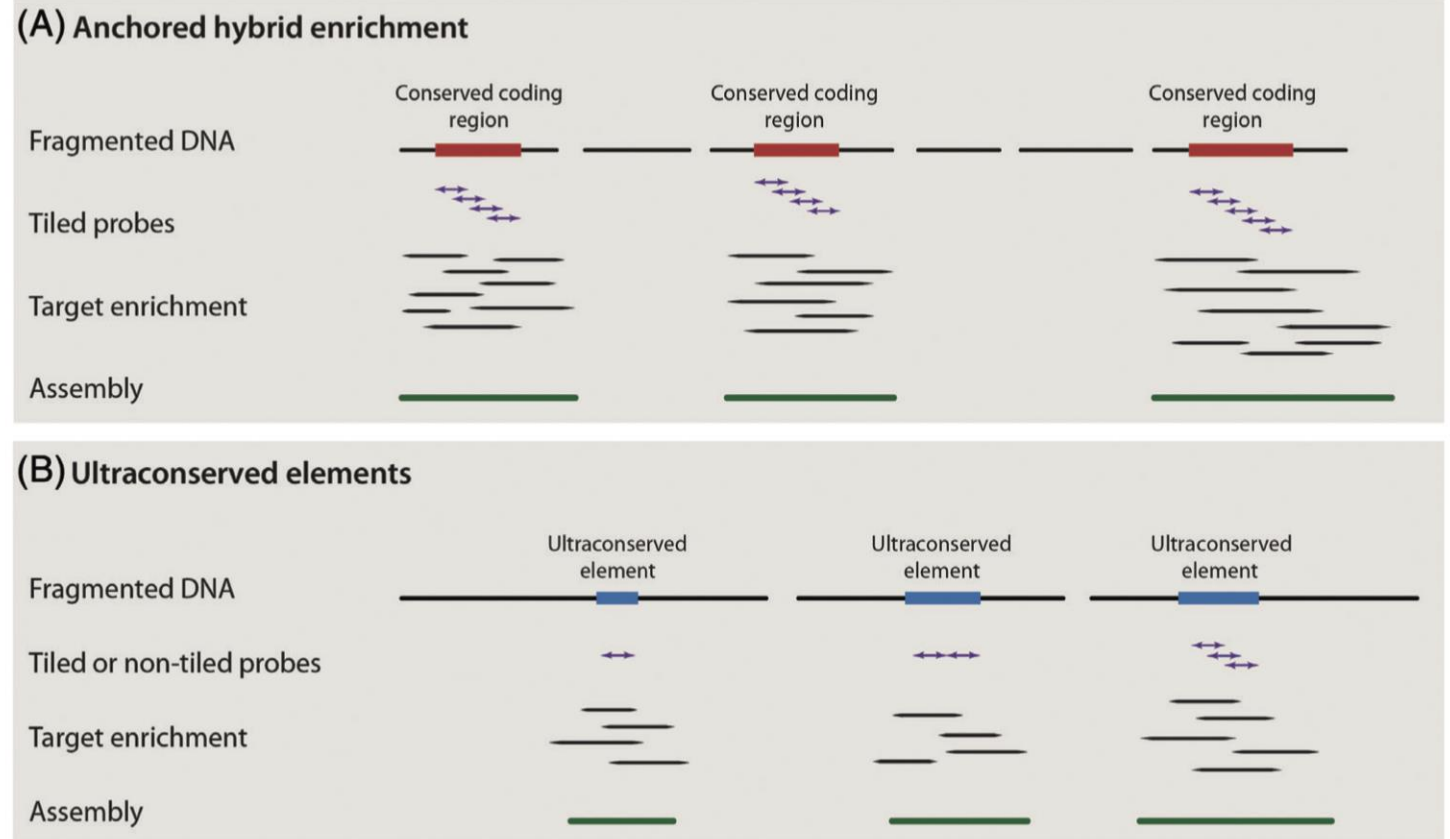


(C) RNA sequencing



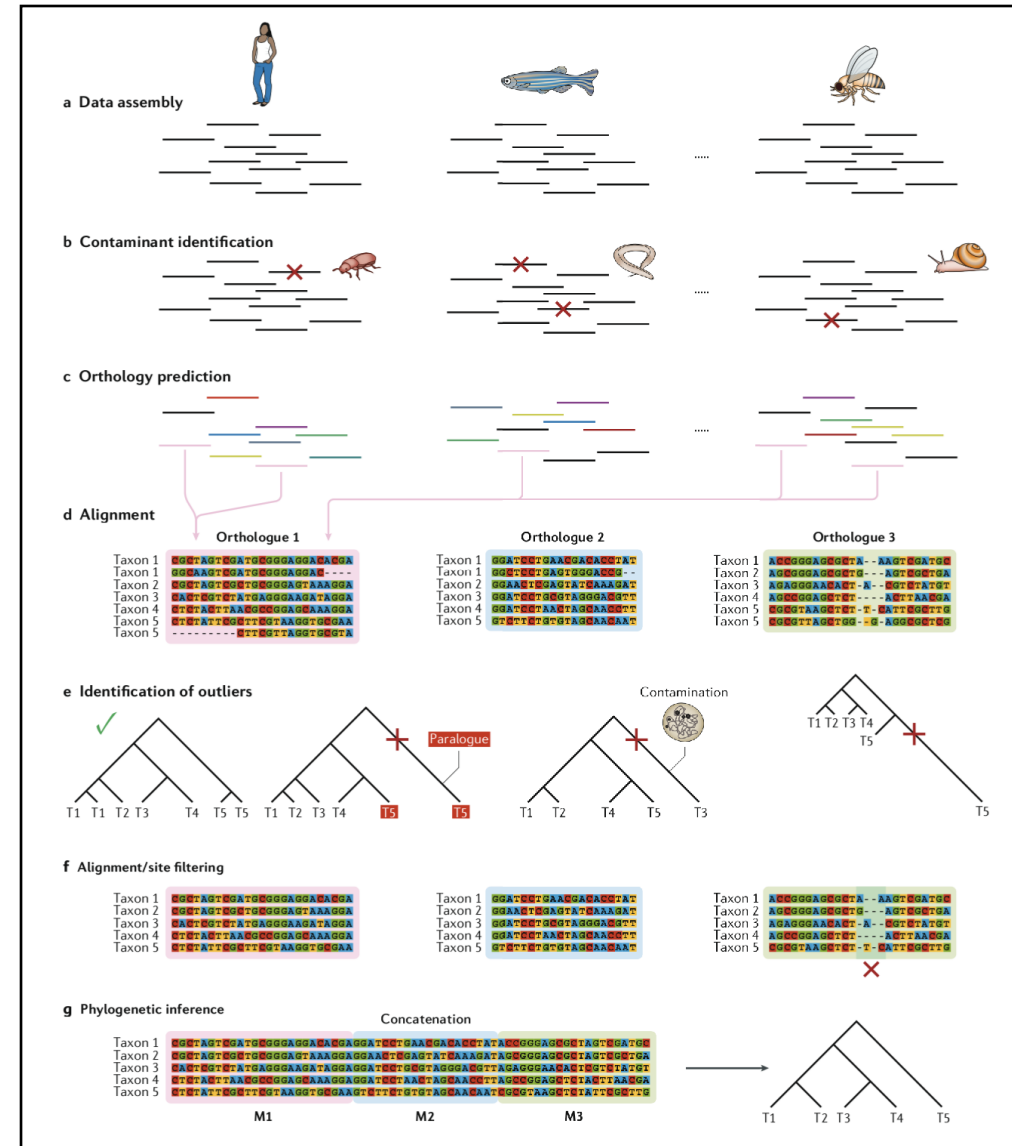
To generate data

- **Target enrichment sequencing:** efficient, cost-effective methods for generating phylogenetic data for nonmodel organisms.
 - **Anchored hybrid enrichment** (mainly targets PCGs of the genome)
 - **Ultraconserved elements** (highly conserved regions of the genome shared among distant taxa of interest)



A pipeline for analyses

- Data assembly
- Contaminant identification
- Orthology prediction
- Alignment
- Identification of outliers
- Alignment/site filtering
- Phylogenetic inference



Orthology prediction

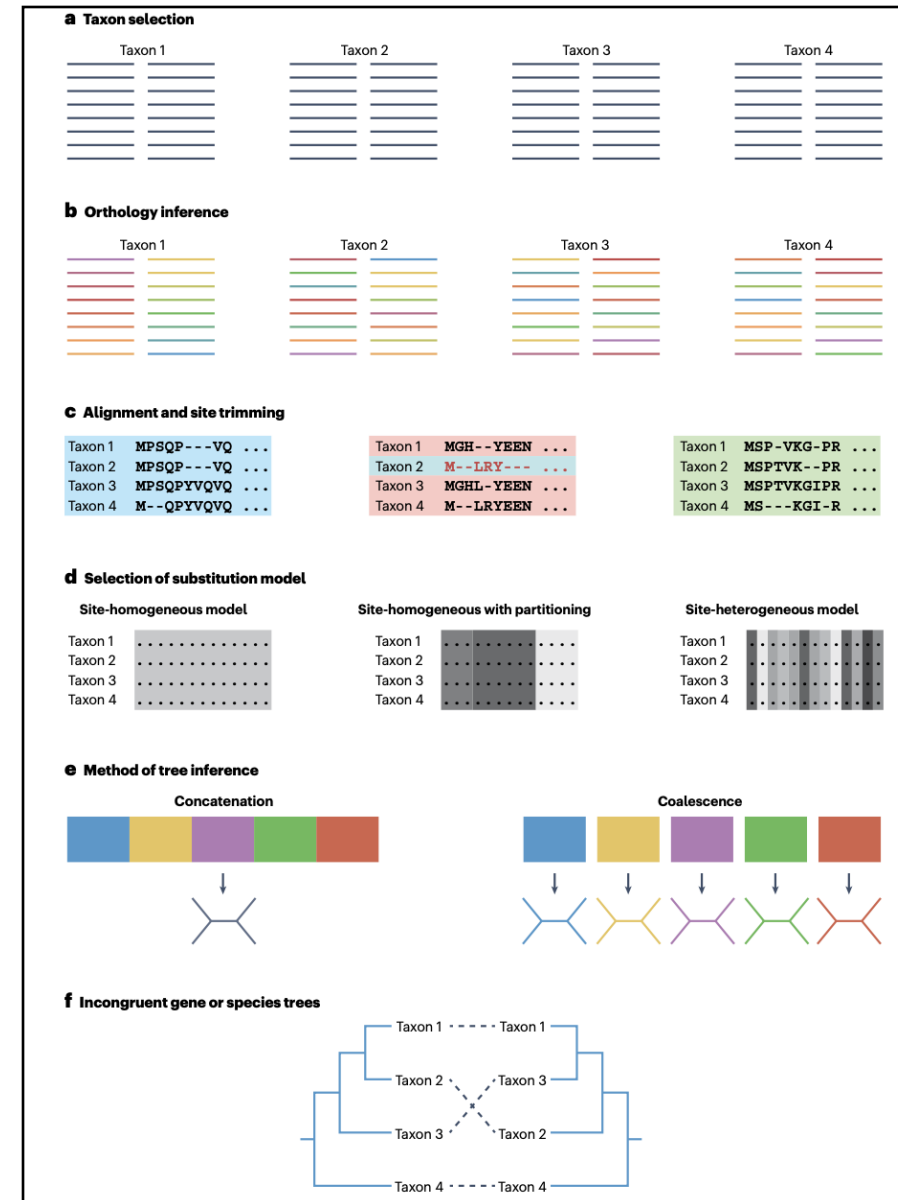
Orthology is a special type of **homology** in which genes in species have diverged from **speciation**.

With the advent of high-throughput data, there is a need of inferring orthologous genes;
tree-based and **graph-based** methods

Method	Type	Comments	Ref.
BUSCO	Graph	Based on precomputed “universal single-copy” genes (defined for a number of standard clades), and thus inherently limited to these. Originally developed to assess genome completeness.	(Waterhouse et al., 2017)
COG/KOG	Graph	One of the first methods, still widely used for prokaryotic data. Includes a manual curation step.	(Tatusov et al., 2003)
EggNOG	Hybrid	Originally developed as extension of COG/KOG. Recent versions also include tree-based refinements.	(Huerta-Cepas et al., 2016b)
ETE 3.0	Tree	General purpose tree analysis and visualisation package for Python, with species overlap function.	(Huerta-Cepas et al., 2016a)

Incongruence of gene or specie trees

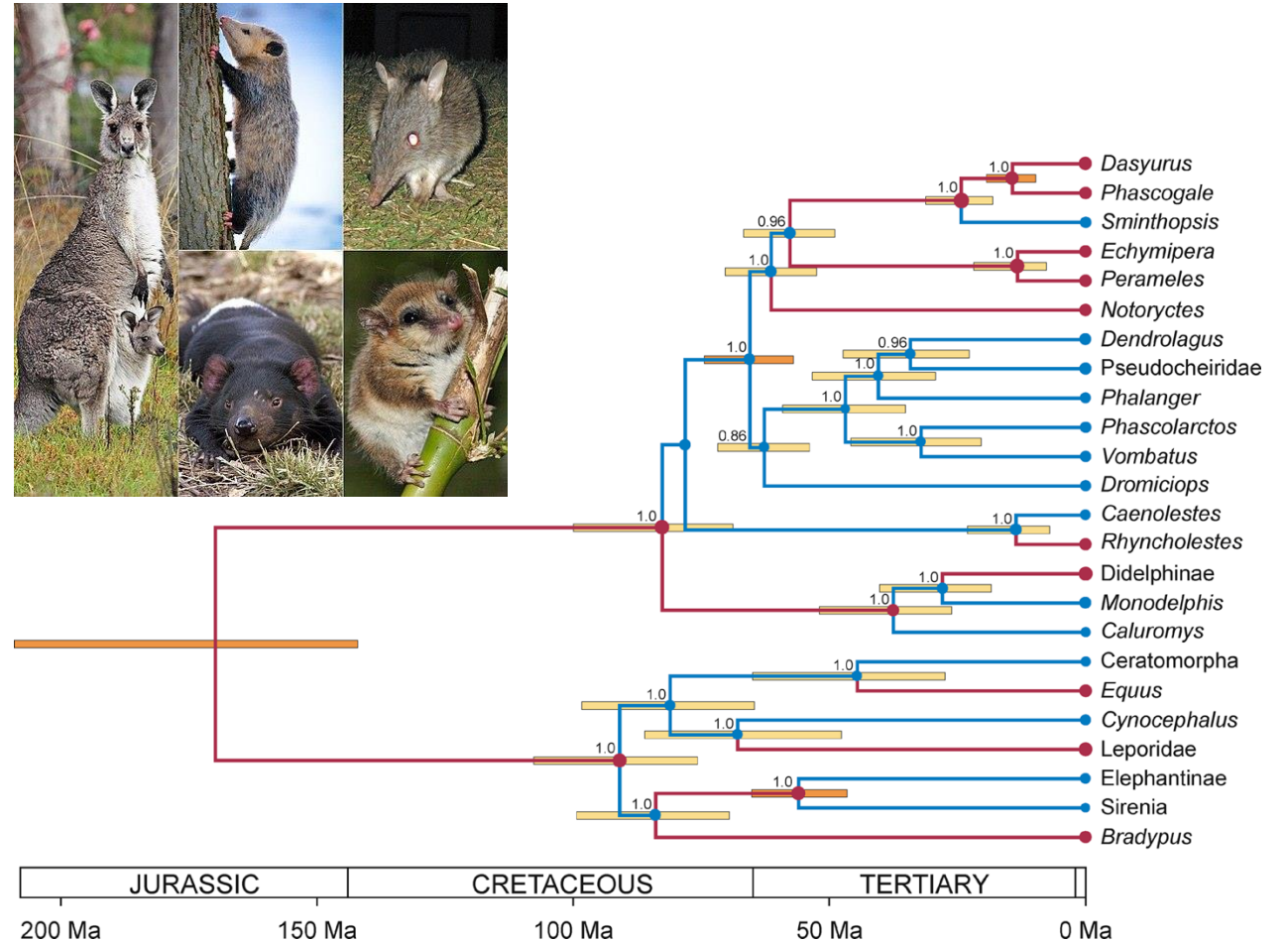
- Despite merits, genome-scale data can sometimes yield conflicting phylogenetic tree topologies
 - **Biological factors** (incomplete lineage sorting, horizontal gene transfer, hybridization, ...)
 - **Analytical factors** (taxon sampling, locus sampling, orthology prediction, alignment trimming, ...)



Application in Phylogenetics and Macroevolutionary Study

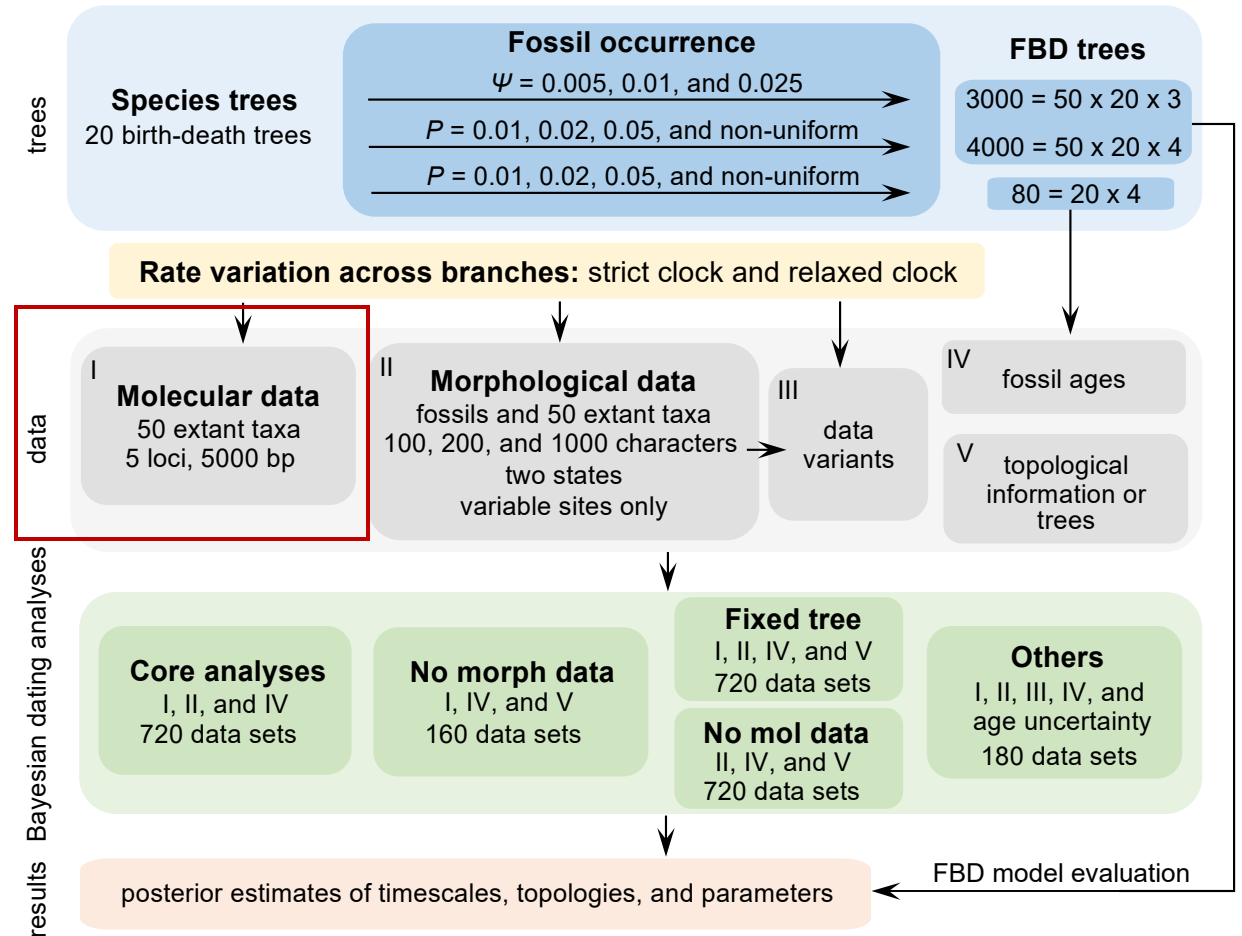
Marsupials and relatives

- 5 PCGs (*APOB*, *RAG1*, *IRBP*, *vWF*, and *BRCA1*) for **17** marsupials and **7** outgroup mammals
- **4** fossil calibration nodes, uncorrelated lognormal model, Yule tree prior, ...
- Co-estimation of phylogeny and divergence under **relaxed molecular-clock modes** with **BEAST 1**
- Simulation and other taxa

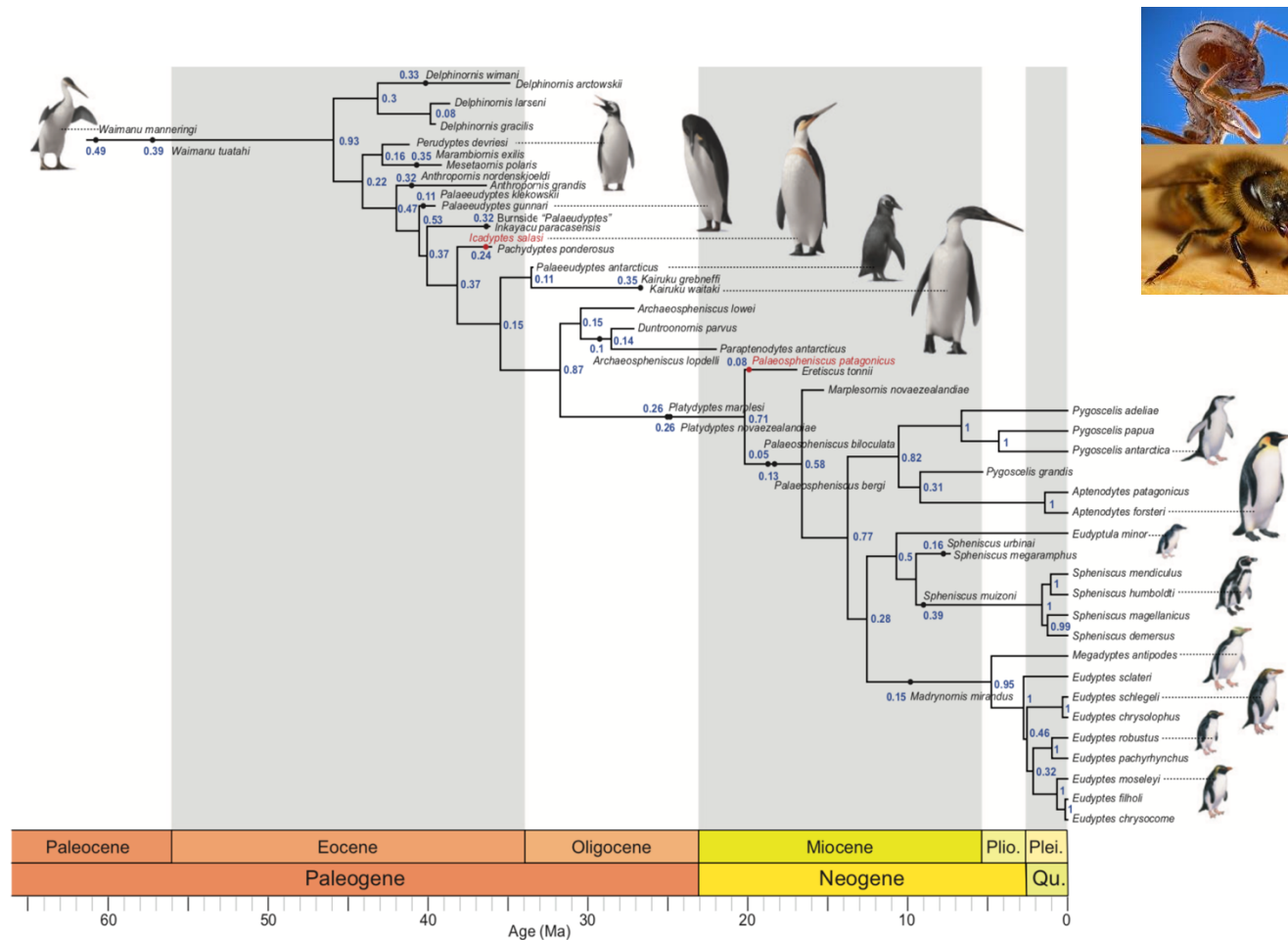


Simulated molecular data

- With full control of the data generating process, **simulated data** help to validate and test hypotheses.
- **50** extant taxa, **5** loci, **1000** bp for each locus of each taxon
- Evaluating the performance of tip-dating using the **fossilized birth-death model**
- **BEAST 2** with the **SA** (sampled ancestor) package

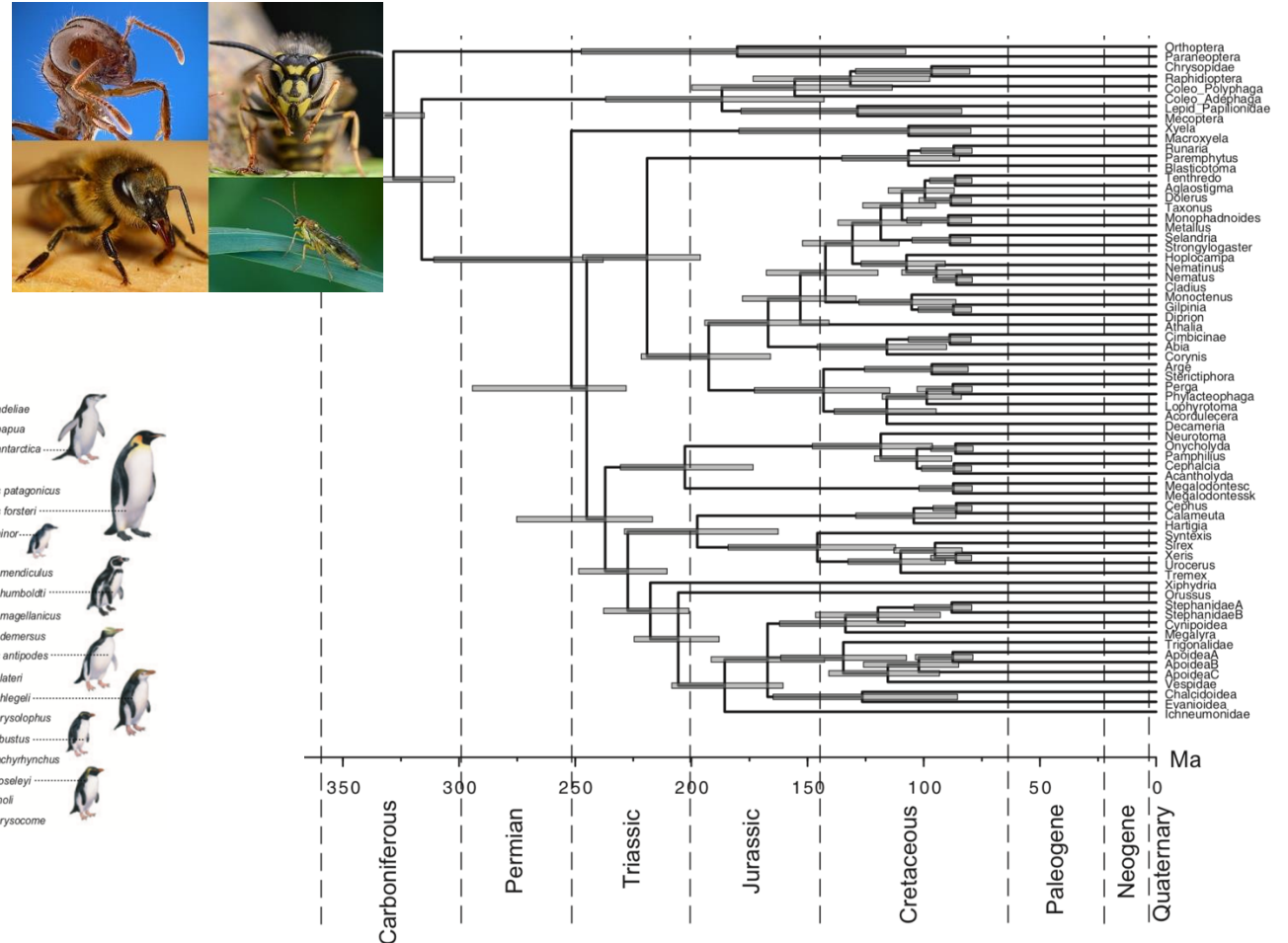


With morphological data



5 genes (8145 bp), 202 characters; 36 fossil species; BEAST2

Gavryushkina *et al.* 2017

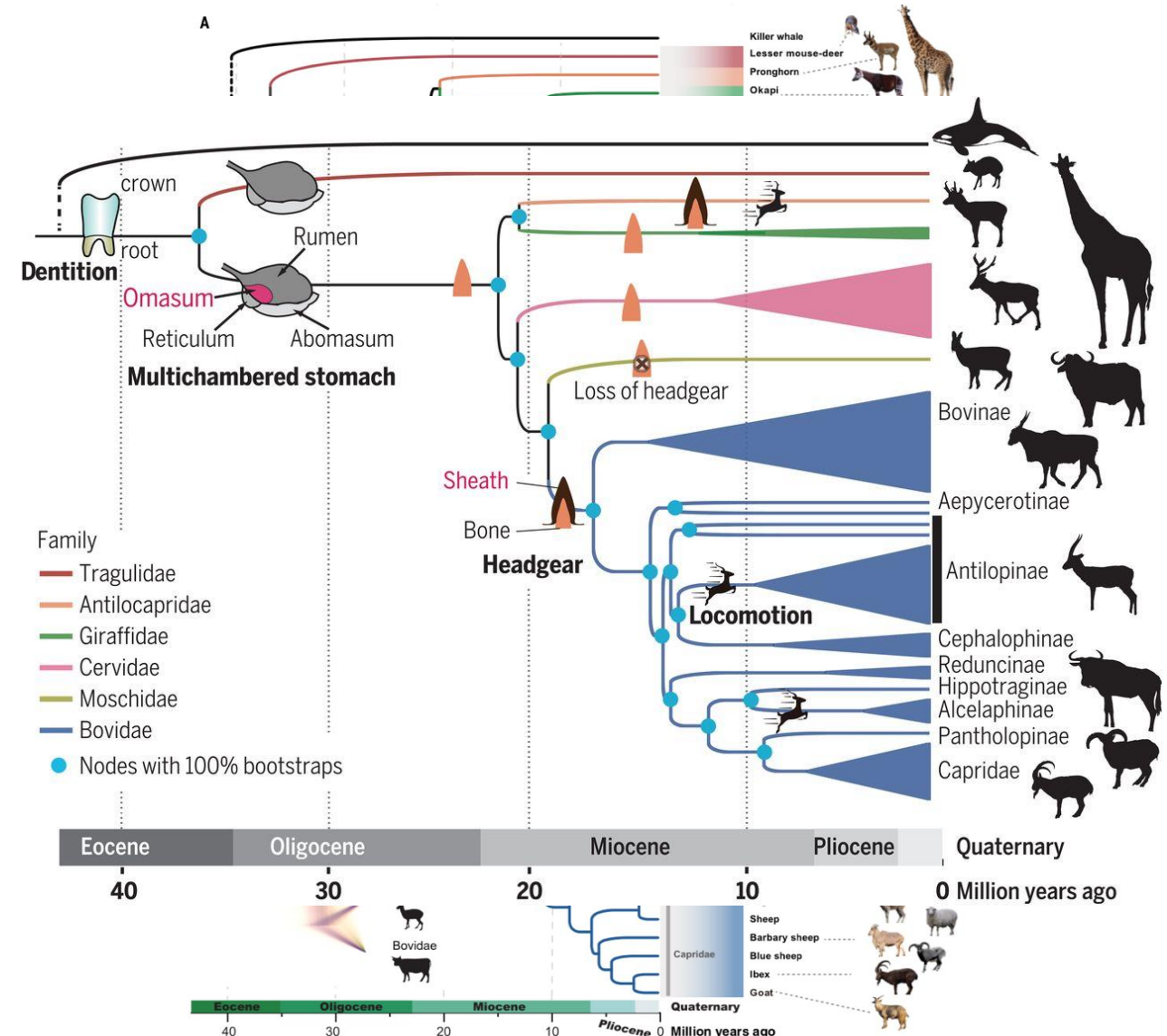


7 genes (5096 bp), 343 characters; 45 fossil species; MrBayes

Zhang *et al.* 2016; Wikipedia

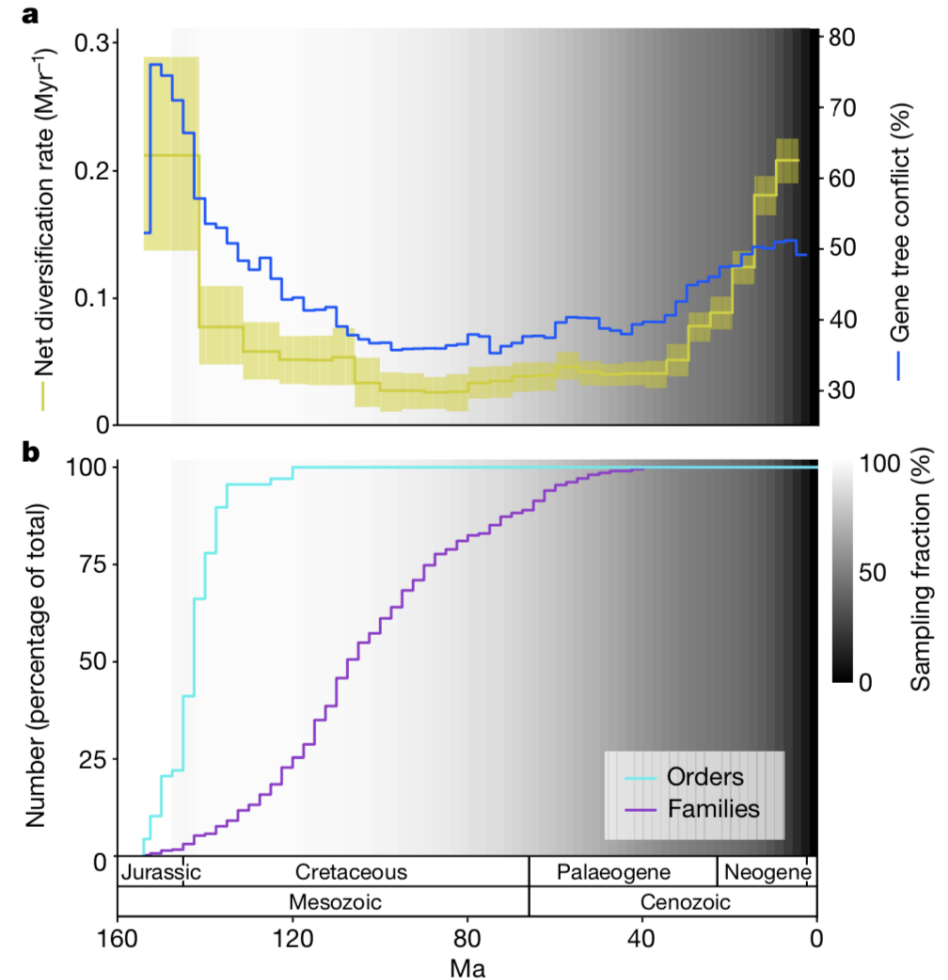
Ruminants

- **51** ruminant species (covering about half the genera and all six extant Ruminantia families)
- **373** Mbp of orthologous syntenic sequences, **6406** orthologous PCGs, the fourfold degenerate sites in these genes, mtDNA, ...
- **Data filtering:** 4d sites of 6 outgroups and 10 ruminant representatives for node dating with **BEAST 1**.
- Evolution of ruminant characteristics (e.g., multichambered stomach)



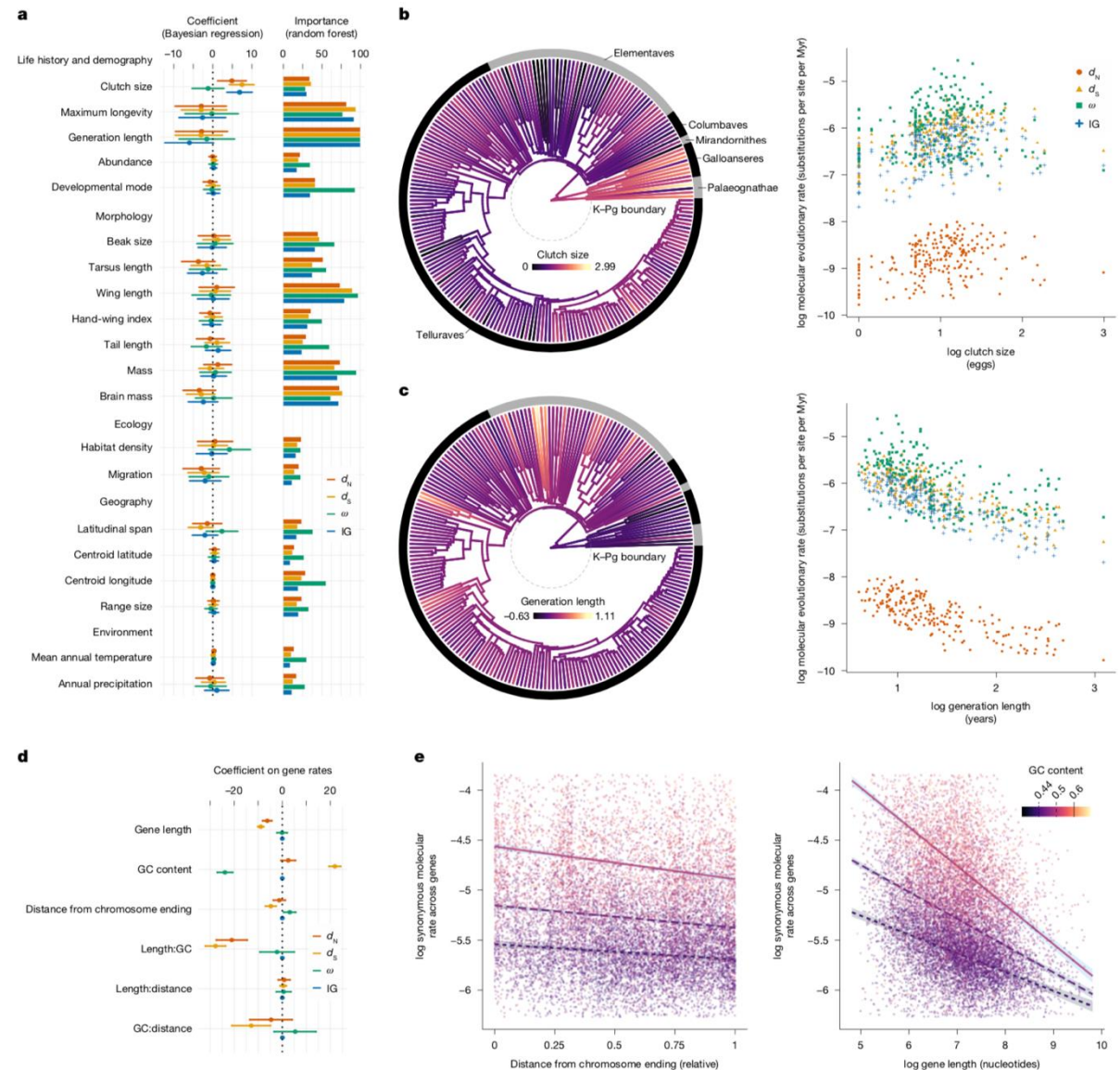
Angiosperms

- **9,506** species (representing 7,923 genera, 416 families, and all 64 orders of angiosperms)
- Target enrichment sequencing; **353** nuclear genes
- Divergence time: selected **25** genes + 200 fossils; penalized likelihood in treePL
- Diversification rate: BAMM and RevBayes
- ...



Modern birds

- **218** species representing all avian families
- **63,174** evenly spaced intergenic loci, **14,849** coding regions of orthologous genes
- **23** traits (including 5 life-history and demography, 7 morphological, 5 ecological, 4 geographical, and 2 environmental traits)
- ...



Summary

- Molecular data forms a fundamental component for phylogenetics and macroevolutionary study of various taxa, which can be of several types, but with sequence data the mostly widely used.
- Several steps for sequence data to be used for phylogenetic analyses: sequence alignment, data partitioning, data filtering, ...
- High-throughput data is now the trend; however, the analyses are more complicated, and sometimes can yield conflicting phylogenetic tree topologies.
- Molecular data can be used together with morphological data, fossils, functional traits, geographic information, etc., to interpret macroevolutionary histories.

Thanks for your attention!

Thanks: Prof. Simon Ho for some slides